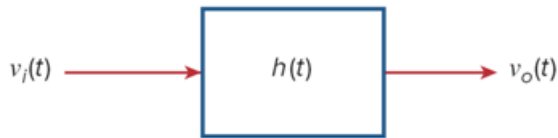


Problem

For the linear system in Fig. 18.54, when the input voltage is $v_i(t) = 2\delta(t)$ V, the output is $v_o(t) = 10e^{-2t} - 6e^{-4t}$ V. Find the output when the input is $v_i(t) = 4e^{-t}u(t)$ V.

Figure 18.54



Step-by-step solution

Step 1 of 2

	INPUT	OUTPUT
Case	$v_i(t) = 2\delta(t)$ V	$v_o(t) = (10e^{-2t} - 6e^{-4t})$ V
	or $v_i(\omega) = 2$	or $v_o(\omega) = \frac{10}{(s+2)} = \frac{6}{(s+4)}$
Case II	$v_i(t) = 4e^{-t}u(t)$ V	$v_o(t) = ?$

now,

$$H(\omega) = \frac{V_o(\omega)}{V_i(\omega)} = \frac{1}{2} \left[\frac{10}{(s+2)} - \frac{6}{(s+4)} \right] = \frac{2(7+s)}{(s+2)(s+4)}$$

Step 2 of 2

now,

$$\begin{aligned} v_o'(\omega) &= v_i'(\omega) H'(\omega) = v_i'(\omega) H(\omega) & [H'(\omega) &= H(\omega)] \\ &= \frac{4}{s+1} \cdot \frac{2(7+s)}{(s+2)(s+4)} = \frac{8(s+7)}{(s+1)(s+2)(s+4)} = \frac{A}{(s+1)} + \frac{B}{(s+2)} + \frac{C}{(s+4)} \end{aligned}$$

On solving this we get,

$$A = 16 \quad B = -20 \quad \text{and} \quad C = 4$$

$$\text{so, } v_o'(\omega) = \frac{16}{(s+1)} - \frac{20}{(s+2)} + \frac{4}{(s+4)}$$

$$\text{or } \boxed{v_o'(t) = (16e^{-t} - 20e^{-2t} + 4e^{-4t})u(t) \text{ V}}$$